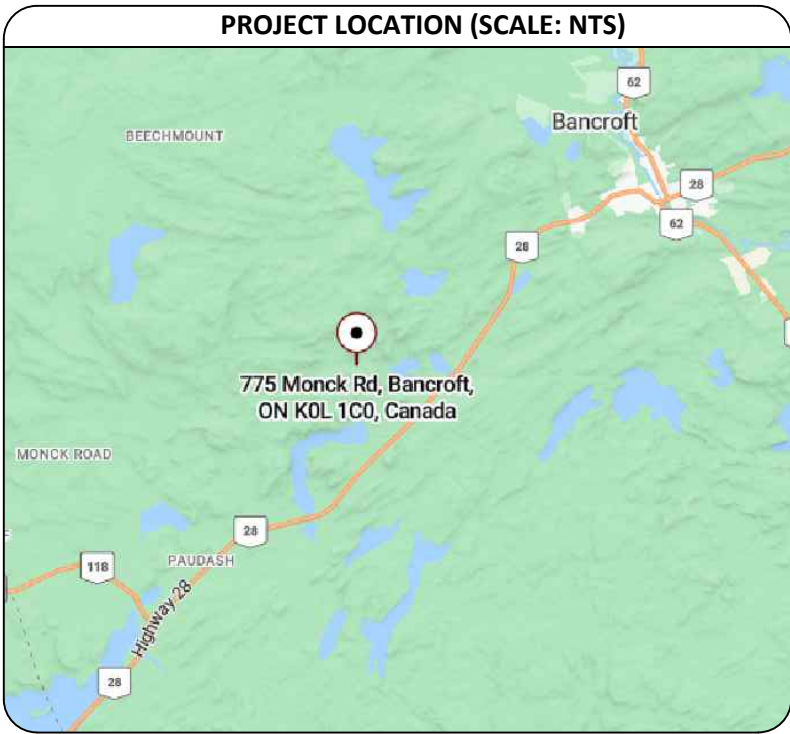


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(18) MODULES x 580W	(1) INVERTER x 7600W
10.440 kW	7.600 kW



SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A RESIDENTIAL LOCATION: 456 STREET NAME, BANCROFT, ON K0L 1C0 . COORDINATES: 45.032820, -77.926980 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM. - INSTALLED PV SYSTEM WILL BE CONNECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ). - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

APPLICABLE GOVERNING CODES & NOTES
THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES: - CANADIAN ELECTRICAL CODE, PART I (CSA C22.1-21, 2021 EDITION) - ONTARIO BUILDING CODE (O. REG. 332/12, AS AMENDED) - ONTARIO FIRE CODE (O. REG. 213/07, AS AMENDED) - LOCAL UTILITY (HYDRO ONE) INTERCONNECTION REQUIREMENTS - APPLICABLE MUNICIPAL BYLAWS (TOWN OF BANCROFT) ALL OTHER RELATED STATE, LOCAL & AHJ AMENDMENTS.

DESIGN CRITERIA
TEMPERATURE SOURCE: ASHRAE WEATHER STATION: WINNIPEG THE FORKS EXTREME MIN. TEMPERATURE: -23 ASHRAE 2% HIGH TEMP: 27

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF ATTACHMENT DETAILS
PV-3	STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
S-2	EQUIPMENT ELEVATION DIAGRAM
E-1 & E-1.1	ELECTRICAL DIAGRAM & CALCULATION
E-2	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

PROJECT INFORMATION	
(#) PV MODULES	(18) THORNOVA SOLAR TS-BGT72 (580W)
(#) INVERTER	(1) FOX ESS AC1-7.6-US & (1) FOX ESS AC1-11.4-US INVERTERS
(#) BATTERY	(1) FOX-ESS ECS4000-H4 (15.9KWH) BATTERY
SYSTEM OUTPUT CURRENT	31.70A
STRING	3 STRINGS OF 06 MODULES
ROOF RACKING SYSTEM	T-FOOT WITH BUTYL SEAL - SUS304 MATERIAL W/ MAPLE LEAF SOLAR MOUNTING RAIL
POINT OF INTERCONNECTION	40A LOAD SIDE BREAKER IN EXTERIOR SUB PANEL WITH 100A MAIN BREAKER, 125A BUSBAR, 120/240V, 3W, 1φ
UTILITY	HYDRO ONE
AHJ	TOWN OF BANCROFT

GENERAL NOTES
- ALL COMPONENTS ARE CSA CERTIFIED, WHERE WARRANTED - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS. - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL READILY ACCESSIBLE. - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH CEC RULE 64-112 & SECTION 10 (GROUNDING AND BONDING) - GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A CSA CERTIFIED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM. - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE. - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS. - WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER CEC 312.2 AND 314.15. - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER CEC 250.92. - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE. - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM. - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION. PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH CSA C22.2 No. 61730 STANDARD.

ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (CEC RULE 64-202) - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONCECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF. - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES. - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL. - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS . - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES. - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT. - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO CEC. - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED. - ALL EXTERIOR CONDUIT TO BE PAINT MATCHED TO THE HOMES COLORS

DATE	--/------
PROJECT ID	OPS
CREATED BY	NSK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE
456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

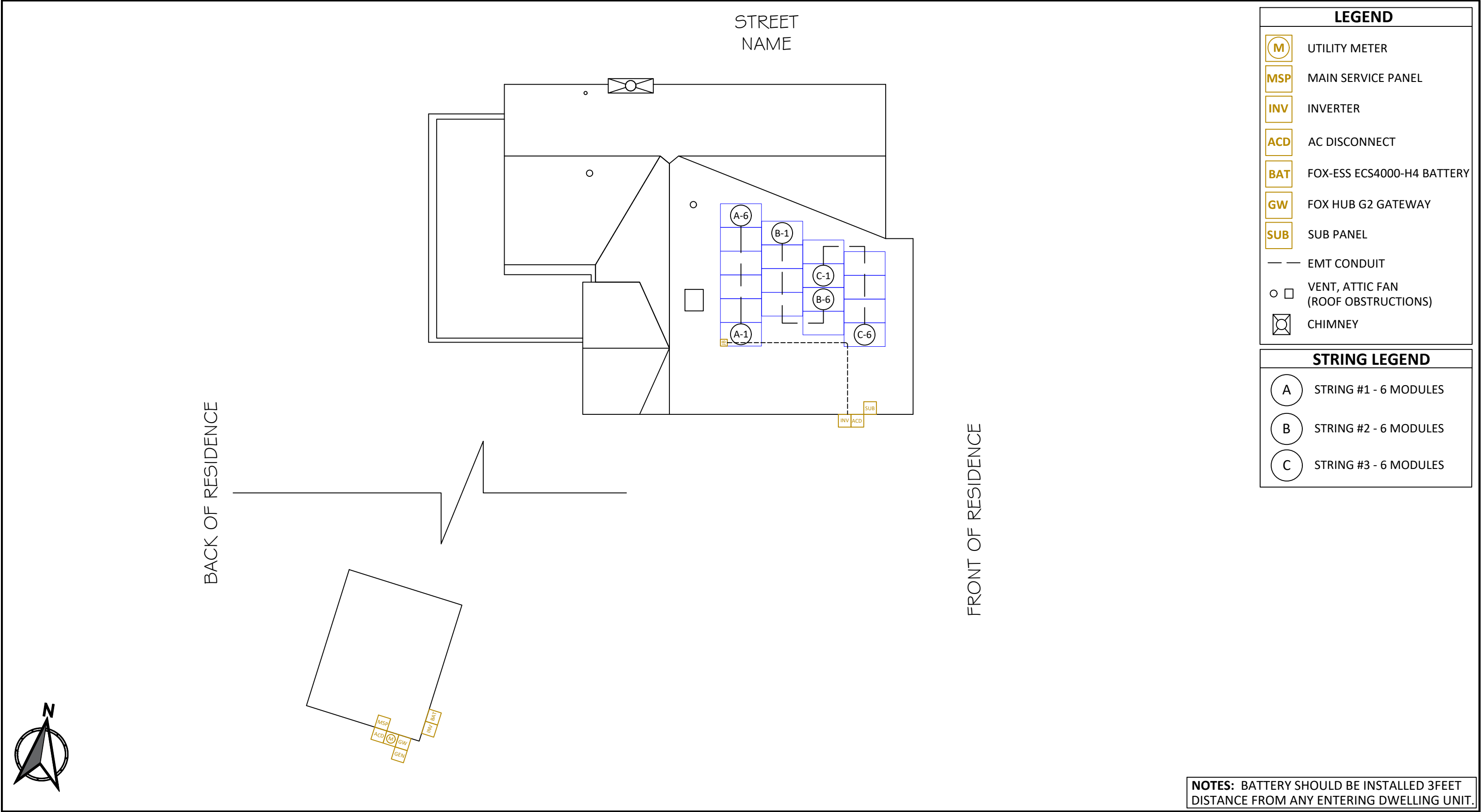
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PV-0

SHEET NO.

PROJECT SUMMARY

SHEET INFORMATION



LEGEND

M

UTILITY METER

MSP

MAIN SERVICE PANEL

INV

INVERTER

ACD

AC DISCONNECT

BAT

FOX-ESS ECS4000-H4 BATTERY

GW

FOX HUB G2 GATEWAY

SUB

SUB PANEL

— —

EMT CONDUIT

○ □

VENT, ATTIC FAN
(ROOF OBSTRUCTIONS)

⊠

CHIMNEY

STRING LEGEND

A

STRING #1 - 6 MODULES

B

STRING #2 - 6 MODULES

C

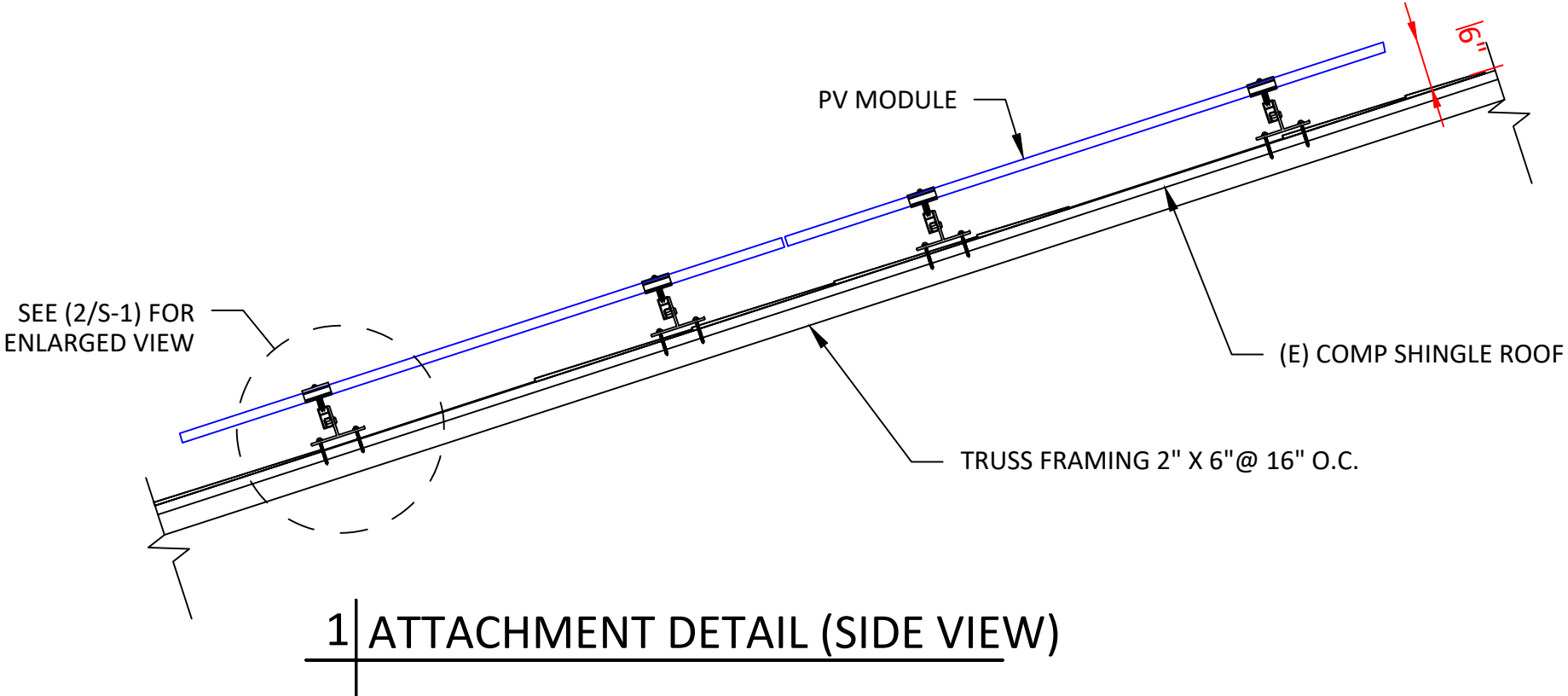
STRING #3 - 6 MODULES

NOTES: BATTERY SHOULD BE INSTALLED 3FEET
DISTANCE FROM ANY ENTERING DWELLING UNIT.

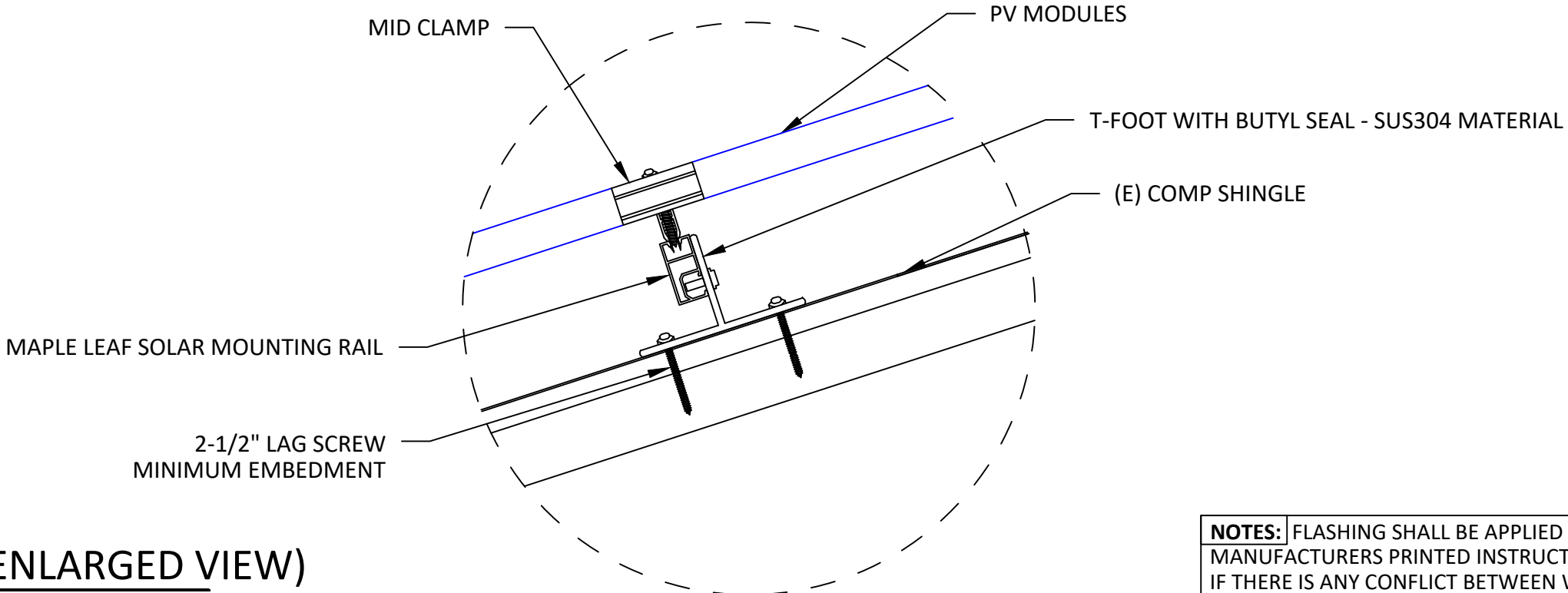
	DATE	--/------																				
	PROJECT ID	OPS																				
	CREATED BY	NSK																				
	CHECKED BY	MSB																				
	SCALE	1/8" = 1'-0"																				
	SHEET SIZE	11"X 17"																				
ANSI B																						
CLIENT RESIDENCE 456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA		<table><tr><th>#</th><th>REVISION/UPDATE</th><th>DATE</th></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr></table>	#	REVISION/UPDATE	DATE																SIGNED & STAMP BY	PV-3 SHEET NO. STRING LAYOUT
			#	REVISION/UPDATE	DATE																	
HOMEOWNER INFO	SHEET INFORMATION																					

STRUCTURAL NOTES:

1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE MAPLE LEAF SOLAR MOUNTING RAIL SYSTEM WITH T-FOOT WITH BUTYL SEAL - SUS304 MATERIAL ATTACHMENTS.
2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
3. THE PROPOSED PV SYSTEM ADDS 2.58 PSF TO THE ROOF FRAMING SYSTEM.
4. ROOF LIVE LOAD = 20 PSF
5. EXPOSURE CATEGORY = B
6. RISK CATEGORY = II



1 | ATTACHMENT DETAIL (SIDE VIEW)



NOTES: FLASHING SHALL BE APPLIED IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.
IF THERE IS ANY CONFLICT BETWEEN WHAT IS DEPICTED & INSTRUCTION AS PER MANUFACTURER, MANUFACTURERS INSTRUCTIONS SHALL SUPERSEDE.

DATE	--/------
PROJECT ID	OPS
CREATED BY	NSK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT RESIDENCE

456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA

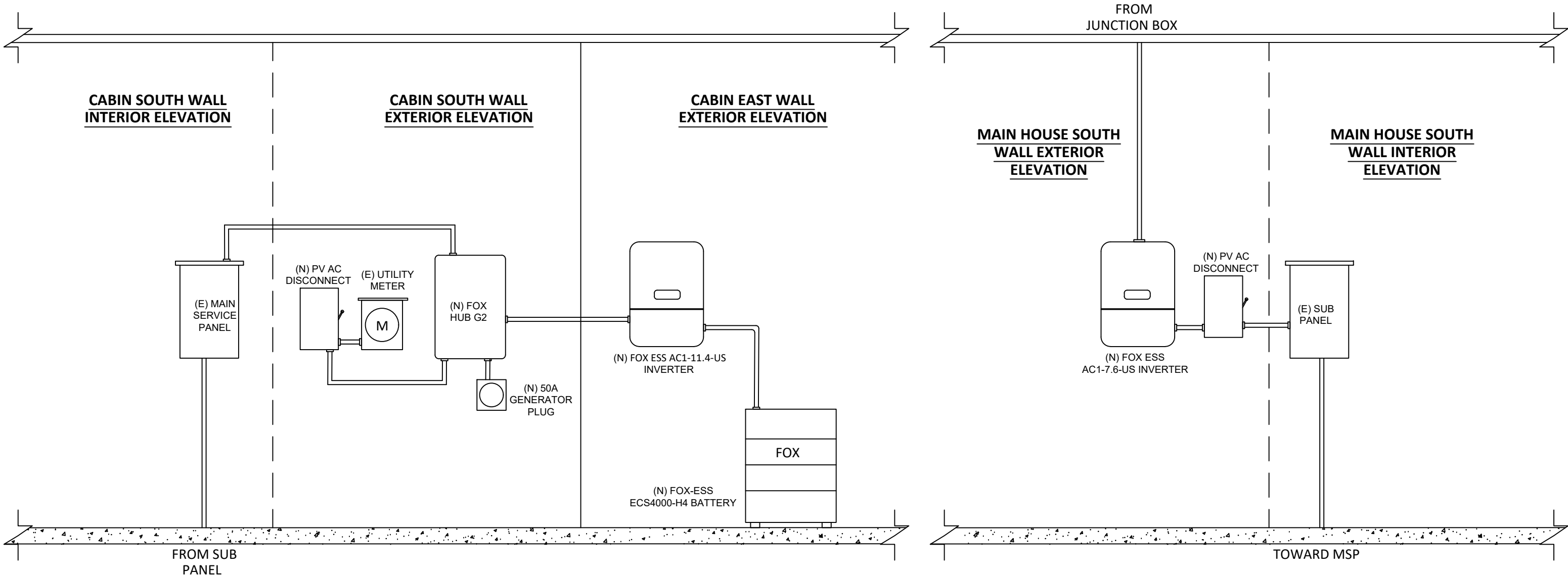
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S-1

SHEET NO.

SHEET INFORMATION

NOTES: BATTERY SHOULD BE INSTALLED 3FEET
DISTANCE FROM ANY ENTERING DWELLING UNIT.



EQUIPMENT ELEVATION DIAGRAM

DATE	--/------
PROJECT ID	OPS
CREATED BY	NSK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE
456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

S-2
SHEET NO.

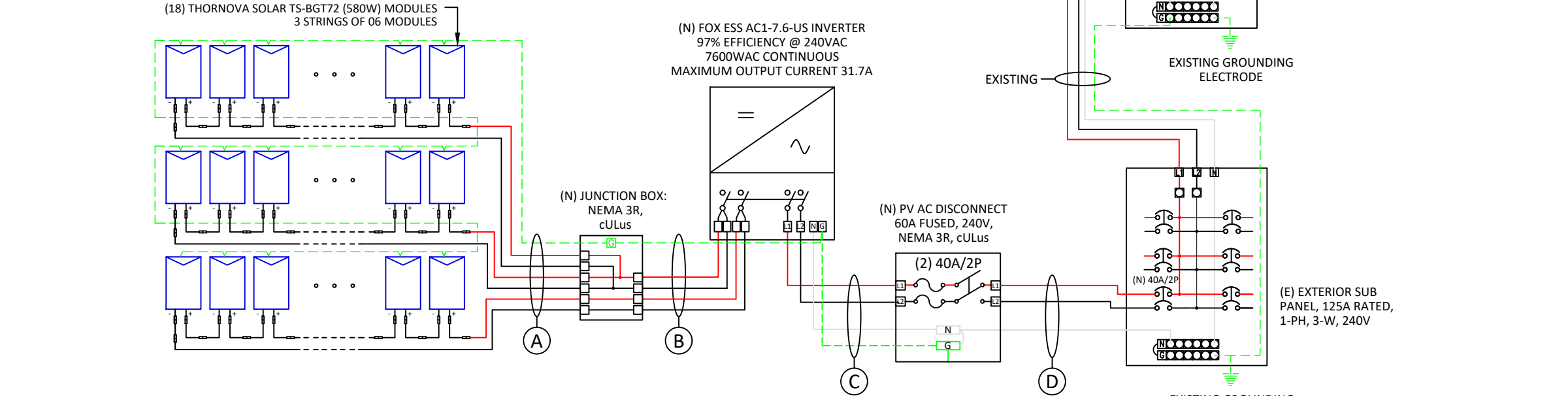
EQUIPMENT
ELEVATION
DIAGRAM

SHEET INFORMATION

EQUIPMENT DETAILS	
(#) PV MODULES	(18) THORNOVA SOLAR TS-BGT72 (580W)
(#) INVERTER	(1) FOX ESS AC1-7.6-US & (1) FOX ESS AC1-11.4-US INVERTERS
(#) BATTERY	(1) FOX-ESS ECS4000-H4 BATTERY
SYSTEM OUTPUT CURRENT	31.70A
POINT OF INTERCONNECTION	40A LOAD SIDE BREAKER IN EXTERIOR SUB PANEL WITH 100A MAIN BREAKER, 125A BUSBAR, 120/240V, 3W, 1φ
UTILITY	HYDRO ONE
AHJ	TOWN OF BANCROFT

SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	THORNOVA SOLAR TS-BGT72 (580W)
VMP	44.04V
IMP	13.17A
VOC	51.97V
ISC	13.80A
TEMP. COEFF. VOC	-0.28 %/°C
MODULE DIMENSION	89.69" (L) x 44.65" (W)
PANEL WATTAGE	580W

PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(18) MODULES x 580W	(1) INVERTER x 7600W
10.440 kW	7.600 kW
120% RULE CEC 64-112(4) (BUS BAR RATING X 120%) - MAIN BREAKER RATING = MAX. PV OCPD (200A x 120%) - 200A = 40A	
PV BREAKER INSTALLED PER CEC 64-112(4) =40A	



NOTES: BATTERY SHOULD BE INSTALLED 3FEET DISTANCE FROM ANY ENTERING DWELLING UNIT.

BATTERY SPECIFICATIONS	
MANUFACTURER / MODEL	FOX-ESS ECS4000-H4
OPERATING TEMPERATURE RANGE	-10°C TO 55°C
USABLE CAPACITY	15.9 KWH

INVERTER SPECIFICATIONS	
MANUFACTURER / MODEL	FOX ESS AC1-11.4-US
NOMINAL OUTPUT VOLTAGE	240VAC
CONTINUOUS OUTPUT CURRENT	47.5A

INVERTER SPECIFICATIONS	
MANUFACTURER / MODEL	FOX ESS AC1-7.6-US
NOMINAL OUTPUT VOLTAGE	240VAC
CONTINUOUS OUTPUT CURRENT	31.7A

DATE	--/------
PROJECT ID	OPS
CREATED BY	NSK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE	
456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA	
HOMEOWNER INFO	

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-1
SHEET NO.
ELECTRICAL DIAGRAM & CALCULATIONS
SHEET INFORMATION

ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT (INVERTER CONTINUOUS CURRENT x 1.25)				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP																																					
A	10 AWG	PV WIRE		OPEN AIR	1	2	N/A	N/A	6 AWG	BARE COPPER	0.82	(48°C)	N/A	13.8A	x	1.25	=	17.3A	N/A	N/A	40FT	0.02%																																					
B	10 AWG	RW90	COPPER	MIN 0.75" Dia EMT	3	6	32.04%	N/A	6 AWG	THWN-2, COPPER	1	(26°C)	0.8	13.8A	x	1.25	=	17.3A	40A	40A x 0.8 x 1 = 32A	35FT	0.08%																																					
C	8 AWG	RW90	COPPER	MIN 0.75" Dia EMT	1	3	29.18%	N/A	6 AWG	THWN-2, COPPER	1	(26°C)	1	31.7A	x	1.25	=	39.6A	55A	55A x 1 x 1 = 55A	5FT	0.11%																																					
D	8 AWG	RW90	COPPER	MIN 0.75" Dia EMT	1	3	29.18%	40A	6 AWG	THWN-2, COPPER	1	(26°C)	1	31.7A	x	1.25	=	39.6A	55A	55A x 1 x 1 = 55A	5FT	0.11%																																					
E	6 AWG	RW90	COPPER	MIN 0.75" Dia EMT	1	3	36.53%	N/A	6 AWG	THWN-2, COPPER	1	(26°C)	1	40.0A	x	1.25	=	50.0A	75A	75A x 1 x 1 = 75A	5FT	0.09%																																					
F	6 AWG	RW90	COPPER	MIN 0.75" Dia EMT	1	3	36.53%	60A	6 AWG	THWN-2, COPPER	1	(26°C)	1	47.5A	x	1.25	=	59.4A	75A	75A x 1 x 1 = 75A	5FT	0.11%																																					
G	3/0 AWG	RW90	COPPER	MIN 1.5" Dia EMT	1	3	39.73%	200A	6 AWG	THWN-2, COPPER	1	(26°C)	1	160.0A	x	1.25	=	200.0A	225A	225A x 1 x 1 = 225A	5FT	0.07%																																					
H	3/0 AWG	RW90	COPPER	MIN 1.5" Dia EMT	1	3	39.73%	200A	6 AWG	THWN-2, COPPER	1	(26°C)	1	160.0A	x	1.25	=	200.0A	225A	225A x 1 x 1 = 225A	5FT	0.07%																																					
<table><tr><td rowspan="6"></td><td>DATE</td><td>--/--/----</td></tr><tr><td>PROJECT ID</td><td>OPS</td></tr><tr><td>CREATED BY</td><td>NSK</td></tr><tr><td>CHECKED BY</td><td>MSB</td></tr><tr><td>SCALE</td><td>NTS</td></tr><tr><td>SHEET SIZE</td><td>11"X 17"</td></tr><tr><td colspan="2"></td><td>ANSI B</td></tr></table> CLIENT RESIDENCE 456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA <table><tr><td>HOMEOWNER INFO</td></tr></table> <table><tr><td>#</td><td>REVISION/UPDATE</td><td>DATE</td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table> SIGNED & STAMP BY E-1.1 SHEET NO. ELECTRICAL DIAGRAM & CALCULATIONS SHEET INFORMATION																							DATE	--/--/----	PROJECT ID	OPS	CREATED BY	NSK	CHECKED BY	MSB	SCALE	NTS	SHEET SIZE	11"X 17"			ANSI B	HOMEOWNER INFO	#	REVISION/UPDATE	DATE																		
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! WARNING

**DO NOT TOUCH TERMINALS.
TERMINALS ON BOTH LINE AND LOAD SIDES MAY
BE ENERGIZED IN THE OPEN POSITION**

! WARNING

**IF GROUND FAULT IS INDICATED
ALL NORMALLY GROUNDED
CONDUCTORS MAY BE
UNGROUND AND ENERGIZED**

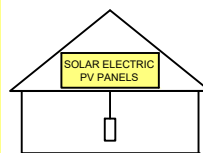
PHOTOVOLTAIC AC DISCONNECT

RATED AC OUTPUT CURRENT	<u>31.70A</u>
NOMINAL OPERATING AC VOLTAGE	240 VAC

RAPID SHUTDOWN SWITCH FOR SOLAR PV SYSTEM

SOLAR PV SYSTEM EQUIPPED WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN
SWITCH TO THE
"OFF" POSITION TO
SHUT DOWN PV SYSTEM
AND REDUCE
SHOCK HAZARD
IN THE ARRAY



PHOTOVOLTAIC SYSTEM UTILITY DISCONNECT SWITCH

PV SOLAR BREAKER

**DO NOT RELOCATE THIS
OVERCURRENT DEVICE**

MAXIMUM VOLTAGE: 480 VDC
MAXIMUM CIRCUIT CURRENT: 15 ADC
MAX. RATED OUTPUT CURRENT OF THE CHARGE CONTROLLER OR DC-TO-DC-CONVERTER (IF INSTALLED) 30 ADC

BATTERY DISCONNECT

**DO NOT RELOCATE
THIS OVERCURRENT
DEVICE**

CAUTION - TRI POWER SOURCE

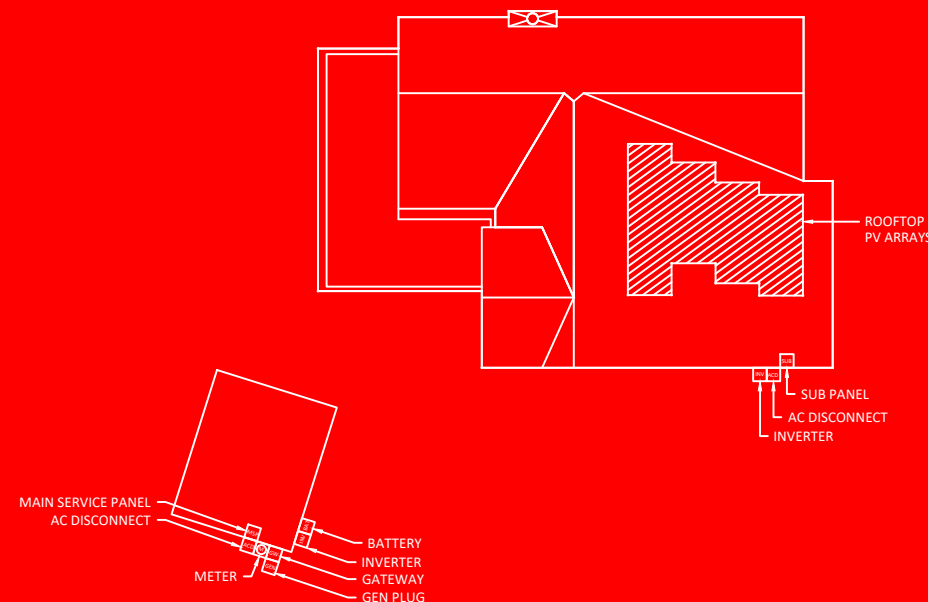
FIRST SOURCE IS UTILITY ELECTRICAL GRID
SECOND SOURCE IS AC BATTERY
THIRD SOURCE IS PV INVERTER

LABEL LOCATION:
SUB PANEL OR BACKUP LOADS PANEL, BATTERY, INVERTER OUTPUT
PANEL, MAIN SERVICE PANEL, METER/MAIN AND PV AC DISCONNECT
CEC RULE 64-218(4)



CAUTION!

MULTIPLE SOURCES OF POWER



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED
CEC Rule 64-218(5)

CLIENT RESIDENCE
456 STREET NAME, BANCROFT, ON K0L 1C0 CANADA

SIGNED & STAMP BY

E-2

SHEET NO.

PLACARDS

SHEET INFORMATION	
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THORNOVA solar

TIER1
BloombergNEF



570-590 W

High efficiency bifacial dual glass module

TS-BGT72



Bifacial technology allows for the harvesting of up to an additional 25% energy from the rear side of the module.



30 years lifespan brings 10-30% additional power generation comparing with conventional P-type module.



N-type solar cell has no LID naturally which can increase power generation.



Excellent low irradiance performance.



Enhanced light trapping and optimized current collection contribute to the improvement of both module power output and reliability.



Industry leading lowest thermal coefficient of power.



Design optimized for lower operating current, resulting in minimized hot spot loss and improved temperature coefficient.



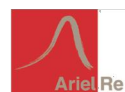
Certified to withstand: wind load (2400 Pa) and snow load (5400 Pa).



100% triple EL test enables remarkable reduction of module hidden crack rate.

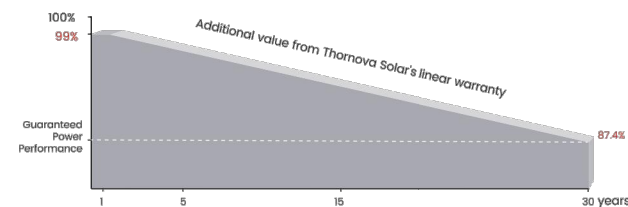
RE INSURANCE

Munich RE



* Optional performance warranty insurance. Please contact our local sales staff for more information.

LINEAR PERFORMANCE WARRANTY



15 years
Product quality & process warranty

30 years
Linear power warranty

0.40 %
Annual degradation
Over 30 years

COMPREHENSIVE CERTIFICATES



ISO 9001: Quality Management System
ISO 14001: Environmental Management System Standard
ISO 45001: International Occupational Health and Safety Assessment System Standard

* Different markets have different certification requirements. Also, the products are under rapid innovation. Please confirm the certification status with regional sales representatives.

ELECTRICAL CHARACTERISTICS

2x72cells

Model of modules	TS-BGT72(570)		TS-BGT72(575)		TS-BGT72(580)		TS-BGT72(585)		TS-BGT72(590)	
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Peak power - P_{mp} (W)	570	436	575	440	580	444	585	448	590	452
Open circuit voltage - V_{oc} (V)	51.52	49.33	51.74	49.54	51.97	49.76	52.16	49.94	52.35	50.12
Short circuit current - I_{sc} (A)	13.70	11.04	13.75	11.08	13.80	11.12	13.85	11.16	13.90	11.20
MPP voltage - V_{mp} (V)	43.62	41.77	43.83	41.97	44.04	42.17	44.22	42.34	44.40	42.51
MPP current - I_{mp} (A)	13.07	10.45	13.12	10.49	13.17	10.53	13.23	10.58	13.29	10.63
Module efficiency - η_m (%)	22.1		22.3		22.5		22.6		22.8	

STC (Standard Testing Conditions): Irradiance 1000W/m², Cell Temperature 25 °C, Spectra at AM1.5

NOCT (Nominal Operating Cell Temperature): Irradiance 800W/m², Ambient Temperature 20°C, Spectra at AM1.5, Wind at 1m/s

ELECTRICAL CHARACTERISTICS WITH DIFFERENT POWER BIN (REFERENCE TO 13.5% IRRADIANCE RATIO)

Peak power - P_{mp} (W)	632	637	643	648	654
Open circuit voltage - V_{oc} (V)	51.52	51.74	51.97	52.16	52.35
Short circuit current - I_{sc} (A)	15.18	15.24	15.29	15.35	15.40
MPP voltage - V_{mp} (V)	43.62	43.83	44.04	44.22	44.40
MPP current - I_{mp} (A)	14.48	14.54	14.59	14.66	14.72
Irradiance ratio (rear/front)	13.5 %				

STRUCTURAL CHARACTERISTICS

Module dimension (L*W*H)	2278 x 1134 x 35 mm (89.69 x 44.65 x 1.38 inch)
Weight	31.5 kg (69.45 lbs)
Number of cells	144 cells
Cell	N-type monocrystalline
Glass	(F)2.0mm, Anti-Reflection Coating (B)2.0mm, Heat Strengthened Glass
Frame	Anodized aluminum alloy
Junction box	IP68, 3 bypass diodes
Output wire	4.0 mm ²
Wire length	300 mm / 1200 mm / Customized length
Connector	MC4 - EVO2
Packing specification	31 pcs/Pallet; 558 pcs/40'HQ

OPERATING PARAMETERS

Power tolerance (W)	(0,+5)
Maximum system voltage (V)	1500
Maximum rated fuse current (A)	30
Current operating temperature (°C)	-40~+85 °C
Bifaciality	80±5%

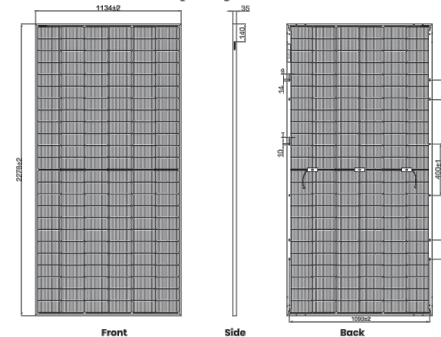
MECHANICAL LOADING

Front side maximum static loading (Pa)	5400
Rear side maximum static loading (Pa)	2400
Hailstone test (mm)	40

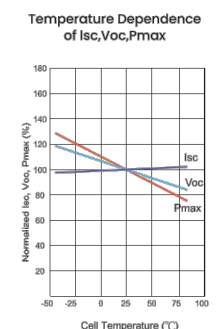
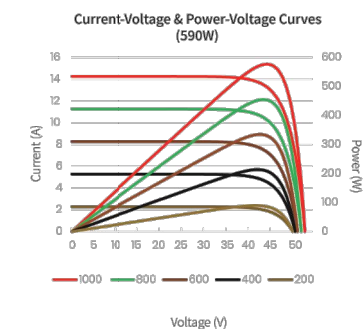
TEMPERATURE RATINGS

Temperature coefficient (P_{max})	-0.29 %/°C
Temperature coefficient (V_{oc})	-0.28 %/°C
Temperature coefficient (I_{sc})	+0.04 %/°C
Nominal operating cell temperature	45±2 °C

MODULE DIMENSIONS (MM)



* The unmarked tolerance is ±1 mm
Length shown in mm



Scan the QR code to
get more information

Web: www.thornovasolar.com

E-mail: info@thornovasolar.com

* The parameters delineated within this datasheet, both technical and monetary, may exhibit variations contingent upon the region. Thornova Solar provides no warranty as to their absolute accuracy. Owing to our unceasing commitment to innovation, research, development, and product enhancement, Thornova Solar retains the discretion to amend any information encapsulated in this datasheet without any preceding notification. Clients are urged to procure the most recent iteration of this datasheet and incorporate it as an intrinsic component of the legally binding agreement ratified by both parties. The English rendition of this datasheet serves purely as a point of reference. Should discrepancies arise between the English text and versions rendered in other languages, the stipulations of the English version shall take precedence.

www.thornovasolar.com

Made in Indonesia

v2024.8.14

THORNOVA solar

TIER1
BloombergNEF



AUTHORIZATION TO MARK

This authorizes the application of the Certification Mark(s) shown below to the models described in the Product(s) Covered section when made in accordance with the conditions set forth in the Certification Agreement and Listing Report. This authorization also applies to multiple listee model(s) identified on the correlation page of the Listing Report.

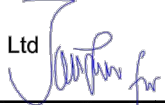
This document is the property of Intertek Testing Services and is not transferable. The certification mark(s) may be applied only at the location of the Party Authorized To Apply Mark.

Applicant: PT THORNOVA SOLAR INDONESIA **Manufacturer:** Yuncheng Solar Technology Co., Ltd

Address: Jalan Brigjen Katamso, Kel. Tanjung
Uncang, Kec. Batu Aji, Kota Batam,
Provinsi Kepulauan Riau, 29466 **Address:** Building H and Building E (production),
Phase II of Standard Plant, Runzhou
Road, Huishan Industrial
Transformation Cluster, Wuxi, Jiangsu

Country: Indonesia **Country:** China

Party Authorized To Apply Mark: Same as Manufacturer
Report Issuing Office: Intertek Testing Services (Shanghai FTZ) Co., Ltd

Control Number: 5026027 **Authorized by:** 
for L. Matthew Snyder, Certification Manager



This document supersedes all previous Authorizations to Mark for the noted Report Number.

This Authorization to Mark is for the exclusive use of Intertek's Client and is provided pursuant to the Certification agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this Authorization to Mark. Only the Client is authorized to permit copying or distribution of this Authorization to Mark and then only in its entirety. Use of Intertek's Certification mark is restricted to the conditions laid out in the agreement and in this Authorization to Mark. Any further use of the Intertek name for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. Initial Factory Assessments and Follow up Services are for the purpose of assuring appropriate usage of the Certification mark in accordance with the agreement, they are not for the purposes of production quality control and do not relieve the Client of their obligations in this respect.

Intertek Testing Services NA Inc.
545 East Algonquin Road, Arlington Heights, IL 60005
Telephone 800-345-3851 or 847-439-5667 Fax 312-283-1672

AUTHORIZATION TO MARK

Photovoltaic (PV) Module Safety Qualification - Part 1: Requirements for Construction [UL 61730-1:2022 Ed.2]
Photovoltaic (PV) Module Safety Qualification - Part 2: Requirements for Testing [UL 61730-2:2022 Ed.2+R:25Apr2023]
Photovoltaic (PV) Module Safety Qualification - Part 1: Requirements for Construction [CSA C22.2#61730-1:2019 Ed.2]
Standard(s): Photovoltaic (PV) Module Safety Qualification - Part 2: Requirements for Testing [CSA C22.2#61730-2:2019 Ed.2]
Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1: Test Requirements [UL 61215-1:2021 Ed.2]
Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1-1: Special Requirements for Testing of Crystalline Silicon Photovoltaic (PV) Modules [UL 61215-1-1:2021 Ed.2]
Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 2: Test Procedures [UL 61215-2:2021 Ed.2]

Product: Crystalline Silicon Photovoltaic (PV) Modules

Brand Name: SUNOVA SOLAR, THORNOVA SOLAR

Models:
TS-BG78(followed by 575, 580, 585, 590, 595, 600, 605 or 610; followed by) or)-X.
TS-BG72(followed by 530, 535, 540, 545, 550, 555, 560 or 565; followed by) or)-X.
TS-BG66(followed by 485, 490, 495, 500, 505, 510 or 515; followed by) or)-X.
TS-BG60(followed by 440, 445, 450, 455, 460, 465 or 470; followed by) or)-X.
TS-BG54(followed by 400, 405, 410, 415, 420 or 425; followed by) or)-X.
TS-BG72(followed by 440, 445, 450, 455 or 460; followed by).
TS-BG60(followed by 365, 370, 375, 380 or 385; followed by).
TS-BG54(followed by 330, 335, 340 or 345; followed by) or).
TS-BGT54(followed by 420, 425, 430, 435, 440, 445, 450, 455 or 460; followed by) or)-X.
TS-BGT60(followed by 465, 470, 475, 480, 485, 490, 495, 500, 505 or 510; followed by) or)-X.
TS-BGT66(followed by 515, 520, 525, 530, 535, 540, 545, 550, 555 or 560; followed by) or)-X.
TS-BGT72(followed by 560, 565, 570, 575, 580, 585, 590, 595, 600, 605, 610 or 615; followed by) or)-X.
TS-BGT78(followed by 605, 610, 615, 620, 625, 630, 635, 640, 645, 650, 655, 660 or 665; followed by) or)-X.
TS-BGT66(followed by 680, 685, 690, 695, 700, 705, 710, 715, 720, 725 or 730; followed by)-G12 or)-G12-X.
TS-BGT60(followed by 620, 625, 630, 635, 640, 645, 650, 655, 660 or 665; followed by)-G12 or)-G12-X.
TS-BGT48(followed by 435, 440, 445 or 450; followed by)-G11 or)-G11-X.
TS-BGT54(followed by 485, 490, 495, 500 or 505; followed by)-G11 or)-G11-X.
TS-BGT60(followed by 545, 550, 555, 560 or 565; followed by)-G11 or)-G11-X.
TS-BGT66(followed by 595, 600, 605, 610, 615 or 620; followed by)-G11 or)-G11-X.
SS-BG followed by 575, 580, 585, 590, 595, 600, 605 or 610; followed by -78MDH or -78MDH-X.
SS-BG followed by 530, 535, 540, 545, 550, 555, 560 or 565; followed by -72MDH or -72MDH-X.
SS-BG followed by 485, 490, 495, 500, 505, 510 or 515; followed by -66MDH or -66MDH-X.
SS-BG followed by 440, 445, 450, 455, 460, 465 or 470; followed by -60MDH or -60MDH-X.
SS-BG followed by 400, 405, 410, 415, 420 or 425; followed by -54MDH or -54MDH-X.
SS-BG followed by 440, 445, 450, 455 or 460; followed by -72MDH.
SS-BG followed by 365, 370, 375, 380 or 385; followed by -60MDH.
SS-BG followed by 330, 335, 340 or 345; followed by -54MDH.



DATASHEET



US SERIES

SPLIT-PHASE HYBRID/AC



HIGH VOLTAGE

Compatible with high-voltage batteries for maximum round-trip efficiency.



EASY INSTALLATION

Flexible configuration, plug and play set-up.



TYPE 4X

Engineered to last with maximum flexibility. Suitable for outdoor installation.



REMOTE MONITORING

Monitor your system remotely via smartphone app or web portal.



The design of the revolutionary single phase storage inverter is tailored to meet the US market. With wide range of power output capacity and up to 4-unit parallel operation, this Fox ESS US Series offers a variety of applications to all customers.

3.8kW ... 11.4kW



For more information about the Fox ESS range, visit:
us.fox-ess.com

TECHNICAL SPECIFICATIONS

MODEL	AC1-3.8-US H1-3.8-US	AC1-5.7-US H1-5.7-US	AC1-7.6-US H1-7.6-US	AC1-9.6-US H1-9.6-US	AC1-11.4-US H1-11.4-US
PV INPUT (ONLY FOR HYBRID)					
Max. Solar STC Power [W]	7600	11400	15200	19200	22800
Max. Input Voltage [V]			600		
Start-up Input Voltage [V]			100		
Rated Input Voltage [V]			380		
MPPT Operating Voltage Range [V]			80 ~ 550		
MPPT Operating Voltage Range [V] (Full Load)	204 ~ 500	204 ~ 500	271~500	257 ~ 500	305 ~ 500
Max. Input Current [A]	28/14	28/14	28/14	28/14/14	28/14/14
Max. Short-circuit Current [A]	44/22	44/22	44/22	44/22/22	44/22/22
No. of MPP Trackers	2	2	2	3	3
Strings per MPP Tracker	2/1	2/1	2/1	2/1/1	2/1/1
BATTERY CONNECTION					
Battery Type	Lithium Battery (LFP)				
Battery Voltage Range [V]	85 ~ 460				
Rated Battery Voltage [V]	360				
Max. Continuous Charge / Discharge Current [A]	50				
Max. Continuous Charge / Discharge Power [W]	(for H1) 5700/4180	8550/6270	11400/8360	14400/10560	17100/12540
	(for AC1) 3800/4180	5700/6270	7600/8360	9600/10560	11400/12540
Max. Discharge Current (60s) [A]	60				
BMS Communication Interface	CAN 2.0				
AC INPUT AND OUTPUT (GRID)					
Max. AC Input Power [VA]	3800	5700	7600	9600	11400
Max. AC Input Current [A]	16	24	32	40	48
Input Voltage Range [V]	211 ~ 264				
Input Frequency Range [Hz]	57 ~ 63				
Rated Output Power [W]	3800	5700	7600	9600	11400
Max. Output Apparent Power [VA]	3800	5700	7600	9600	11400
Rated Output Current [A]	15.8	23.8	31.7	40.0	47.5
Max. Output Current [A]	15.8	23.8	31.7	40.0	47.5
Rated Grid Voltage [V]	240 (211 ~ 264)				
Rated Grid Frequency [Hz]	57 ~ 63				
Power Factor	>0.99 (Adjustable from 0.8 leading to 0.8 lagging)				
THDi [%]	<3 @Rated power				
AC OUTPUT (BACKUP)					
Rated Output Power [W]	11400				
Rated output power @different batteries [W]	5760 @H2 Battery 8640 @H3 Battery 11400 @H4 Battery 11400 @H5 Battery 1140C @H6 Battery (Only for non US mainland) 1140C @H7 Battery (Only for non US mainland)				
Max. Output Apparent Power [VA]	12540				
Peak Output Apparent Power (10min) [VA]	13680				
Peak Output Apparent Power (60s) [VA]	15390				
Max. Continuous Output Current [A]	52.3				
LRA [A]	160				
Rated Output Voltage [V]	120 / 240				
Rated Output Frequency [Hz]	60				
Load Start Capacity [A] LRA	50 @H2 Battery 75 @H3 Battery 110 @H4 Battery 160 @H5 Battery 160 @H6 Battery (Only for non US mainland) 160 @H7 Battery (Only for non US mainland)				
THDv (Linear Load) [%]	<3 @Rated power				
Imbalance for Split-Phase Loads [%]	100				
EFFICIENCY					
CEC Efficiency [%]	97.00				
Max. Efficiency [%]	97.60				
Max. Battery Discharge Efficiency (BAT to AC) (@full load, 340Vdc) [%]	97.40				
PROTECTION					
Insulation Monitoring	YES				
Residual Current Monitoring	YES				
DC Reverse Polarity Protection	YES				
Anti-islanding Protection	YES				
AC Short-circuit Protection	YES				
AC Overcurrent / Overvoltage Protection	YES				
DC Switch	YES				
SPD	DC: Type II / AC: Type II				
AFCI	YES				
GENERAL DATA					
Dimensions (W*H*D) [inch / mm]	18.2*25.9*9.2 / 462*658*234.5				
Weight [lbs / kg]	83.8 / 38				
Installation	Wall-mounted				
Topology	Transformerless				
Cooling Method	Natural convection				
Noise Emission [dB]	<35				
Max. Operating Altitude [ft / m]	9843 / 3000 (Derating above 6560 / 2000)				
Operating Temperature [°F / °C]	-13 ~ +140 / -25 ~ +60 (Derating above 104 / 40)				
Humidity [%]	0 ~ 100 (No Condensation)				
Protection Degree	Type 4X				
Standby consumption [W]	<25				
Max. number of Inverters (Parallel)	4				
Monitoring Module	WiFi, LAN				
Communication	CAN2.0, RS485, Meter, CT, ISO alarm, SUNSPEC				
Display	LED, App, Website				
Warranty [Year]	12.5				
STANDARD COMPLIANCE (MORE AVAILABLE UPON REQUEST)					
Safety	UL1741 SA, UL 1741 SB, UL1741 CRD, HECO SRD-V2.0, CSA C22.2 No.107.1-16, UL1998, UL16998, Rule 21				
EMC	FCC part15 CLASS B				
Grid Regulation	IEEE1547-2018, IEEE1547a-2020, IEEE1547.1-2020				



Certificate of Compliance

Certificate: 80112280 Master Contract: 302680

Project: 80147980 Date Issued: 2023-01-06

Issued To: FOXESS CO., LTD.
No.939, Jinhai Third Road,
New Airport Industry Area,
Longwan District
Wenzhou, Zhejiang, 325025
China

Attention: Hardy Yang

The products listed below are eligible to bear the CSA Mark shown with adjacent indicators 'C' and 'US' for Canada and US or with adjacent indicator 'US' for US only or without either indicator for Canada only.



Issued by: Henry Huan
Henry Huan

PRODUCTS

CLASS - C5311 09 - POWER SUPPLIES Distributed Generation Power Systems Equipment
CLASS - C5311 89 - POWER SUPPLIES - Distributed Generation Power Systems Equipment - Certified to U.S. Standards

Bi-directional Transformerless Power Conversion Equipment, US Series, Models H1-3.8-US, H1-5.7-US, H1-7.6-US, H1-9.6-US and H1-11.4-US with PV input and Battery input, Grid Support Utility Interactive mode or Stand-alone mode. Wall mounted, permanently connected.

Bi-directional Transformerless Power Conversion Equipment, US Series, Models AC1-3.8-US, AC1-5.7-US, AC1-7.6-US, AC1-9.6-US and AC1-11.4-US with Battery input, Grid Support Utility Interactive mode or Stand-alone mode. Wall mounted, permanently connected.



Certificate: 80112280 Master Contract: 302680
Project: 80147980 Date Issued: 2023-01-06

Notes:
For details related to rating, size, configuration, etc., reference should be made to the CSA Certification Record, or the Descriptive Report.

APPLICABLE REQUIREMENTS

- CSA-C22.2 No.107.1-16 - Power Conversion Equipment
- *UL Sid No. 1741 3rd Edition - Inverters, Converters, Controllers and Interconnection System Equipment For Use With Distributed Energy Resources (Third Edition, Revision Dated September 28, 2021)
- **UL1741 CRD - Non-Isolated EPS Interactive PV Inverters Rated Less Than 30Kva (Dated April 26, 2010)
- **UL 1699B - Photovoltaic (PV) DC Arc-Fault Circuit Protection (First Edition, Dated August 22, 2018)

*Note: Conformity to UL 1741(Third Edition, Revision Dated September 28, 2021) includes compliance with UL 1741 Supplement B and IEEE 1547.1-2020 with the SRDs of IEEE 1547-2018, IEEE 1547a-2020, Hawaiian Electric Co. SRD-V2.0 and Puerto Rico Eletric Power authority (PV) Projects (V3). While the grid support function evaluated according to IEEE 1547.1-2020, the interoperability is verified with IEEE 2030.5-2018 communication protocol.

**Note: The functional safety has been evaluated according to applicable requirement of UL 1998-Edition 3 as required by the product standard.

Notes:

Products certified under Class C531109 have been certified under CSA's ISO/IEC 17065 accreditation with the Standards Council of Canada (SCC). www.scc.ca



Supplement to Certificate of Compliance

Certificate: 80112280 Master Contract: 302680

The products listed, including the latest revision described below, are eligible to be marked in accordance with the referenced Certificate.

Product Certification History

Project	Date	Description
80147980	2023-01-06	Updated report 80112280 to add component Microgrid Interconnect Device, model FOX Hub and inverter AFCI Function
80136757	2022-12-09	Updated report 80112280 to add Puerto Rico Eletric Power authority (PV) Projects (V3).
80112280	2022-07-29	Bi-directional Transformerless Power Conversion Equipment, US Series with Grid Support Utility Interactive mode or Stand-alone mode.(C/US)



DATASHEET

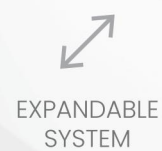


ECS4000

HIGH VOLTAGE STORAGE BATTERY



- 3.97kWh capacity
- Scalable to 27.8kWh
- 90% Depth of Discharge
- Wide Temperature Tolerance
- CAN communication
- Voltage Range up to 459.9V



The ECS is a high-performance, scalable battery storage system. The modular design allows for maximum flexibility, making it suitable for a broad range of storage applications.

Additional batteries can be installed in series, allowing for a maximum storage capacity of 27.8kWh. Installation is easy, with a plug and play solution that can save valuable time for installers.



For more information about the Fox ESS range, visit:
us.fox-ess.com



TECHNICAL SPECIFICATIONS

MODEL	ECS4000-H2	ECS4000-H3	ECS4000-H4	ECS4000-H5	ECS4000-H6	ECS4000-H7
ELECTRICAL CHARACTERISTICS						
Battery Type			LiFePO ₄	Prismatic Cell		
Battery Module	1*CM4000	1*CM4000	1*CM4000	1*CM4000	1*CM4000	1*CM4000
	1*CS4000	2*CS4000	3*CS4000	4*CS4000	5*CS4000	6*CS4000
Nominal Capacity [kWh]	7.95	11.92	15.9	19.87	23.85	27.82
Nominal Voltage [V]	115.2	172.8	230.4	288	345.6	403.2
Operating Voltage [V]	97.2~131.4	145.8~197.1	194.4~262.8	243~328.5	291.6~394.2	340.2~459.9
Max. Charge / Discharge Current [A]				50		
Peak Discharge Current [A]				65 @60sec		
Round-trip Efficiency [%]				>95		
Depth of Discharge [%]				90		
Communication				CAN		
Display				CS: LED*1, CM: LED*6		
Scalability				Max. 7 Modules in Series		
OPERATING CONDITIONS						
Installation Location				Outdoor / Indoor (Stand)		
Operating Temperature [°F / °C]				14 ~ +131 / -10 ~ +55 (Derating above 104 / 40)		
Storage Temperature [°F / °C]				-4 ~ +131 / -20 ~ +55		
Cooling Method				Natural Convection		
Humidity [%]				0 ~ 100 (No condensation)		
Altitude [m / ft]				Max. 2,000 / 6,560		
MECHANICAL CHARACTERISTICS						
Dimensions (W*H*D) [mm]	570*350*380	570*470*380	570*590*380	570*710*380	570*830*380	570*950*380
Dimensions (W*H*D) [inch]	22.4*13.8*15.0	22.4*18.5*15.0	22.4*23.2*15.0	22.4*28.0*15.0	22.4*32.7*15.0	22.4*37.4*15.0
Weight [kg/lbs]	73.5 / 161.7	108.4 / 238.5	143.3 / 315.3	178.2 / 392	213.1 / 468.8	248 / 545.6
CERTIFICATES						
Safety				UL1973, UL9540, UL9540A		
Transportation				UN38.3		
Ingress Protection				IP65		
WARRANTY						
Standard Warranty [year]				12.5*1		

*1, Refer to Fox ESS Battery Warranty Terms and Conditions.



Version 1.2 | 2025/03/11



DATASHEET FOX HUB G2



As a smart energy management system, Fox Hub completes the most critical piece of the puzzle in Fox ESS residential energy storage system. By integrating multiple intelligent features, Fox Hub centralizes power from multiple sources and performs energy redistribution via FoxCloud US App to provide a more comprehensive power solution. It reassures energy flow safety under different circumstances, and relieve homeowners from power loss concern.

- Automatic Off-Grid Protection
- Seamless On/Off Grid Transition
- Smart Load Shedding Circuit
- Inverter Scalability
- Generafor Compatibility



DATASHEET FOX HUB G2

MODULE	Fox Hub G2
GENERAL ELECTRICAL	
AC Current (Nominal) [A]	200
AC Input Voltage (Nominal) [Vac]	240
AC Output Voltage (Nominal) [Vac]	120/240
Rated AC Power [KW]	48
AC Voltage Range [Vac]	211-264
AC Frequency (Nominal) [Hz]	60
AC Frequency Range [Hz]	57-63
Microgrid Interconnection Device Rated Current [A]	200
Grid Disconnection Switchover Time [ms]	<15
EXTERNAL CONNECTION	
3rd Party Solar [A]	80
FOX Inverter [A]	80
Smart Circuits 1, 2 (120V) [A]	80
Smart Circuit 3 (240V) [A]	80
Non-Backup Load [A]	200
Backup Load Terminal [A]	200
GENERATOR (Optional)	
Maximum Rated AC Power [KW]	48
AC Voltage Range [Vac]	211-264
Maximum Continuous Input Current [Adc]	200
Dry Contact Switch Voltage Rating [Vac/Vdc]	250/30
Dry Contact Switch Current Rating [A]	0.5A(250Vac) / 3A(30Vdc)
2-wire Start Switch	Yes
ADDITIONAL FEATURES	
Maximum Number of Inverter	4
Communication	CAN, RS485
Energy Meter (for Import/Export)	0.5% accracy
STANDARD COMPLIANCE	
Safety	UL67,UL869A,UL916,UL1741,CSA22.2 NO.107
Emissions	FCC part 15 class B
INSTALLATION SPECIFICATIONS	
Dimensions (HxWxD) [in / mm]	25.6 in x 17.4 in x 6.2 in /650 mm x 443mm x 156.6 mm
Weight [lb / Kg]	35.3lbs / 16Kg
Supported Inverters	FOX US Series, AIO US Series
Grid Conduit Size / AWG Range	2.5"Conduits / 6AWG - 250kcmil
Generator Input Conduit Size / AWG Range	2.5"Conduits / 6AWG - 250kcmil
Communication Conduit Size / AWG Range	1"Conduits / 24AWG - 16AWG
Cooling	Natural convection
Noise [dBA]	<35dB
Operating Temeptrature Range [°F / °C]	-13°F~+122°F (-25 °C ~ +50 °C)
Protection Rating	Type 3R
Warranty [Years]	12.5

Maple Leaf Black T-Foot with Butyl Seal Functionality



Superior Weather Resistance



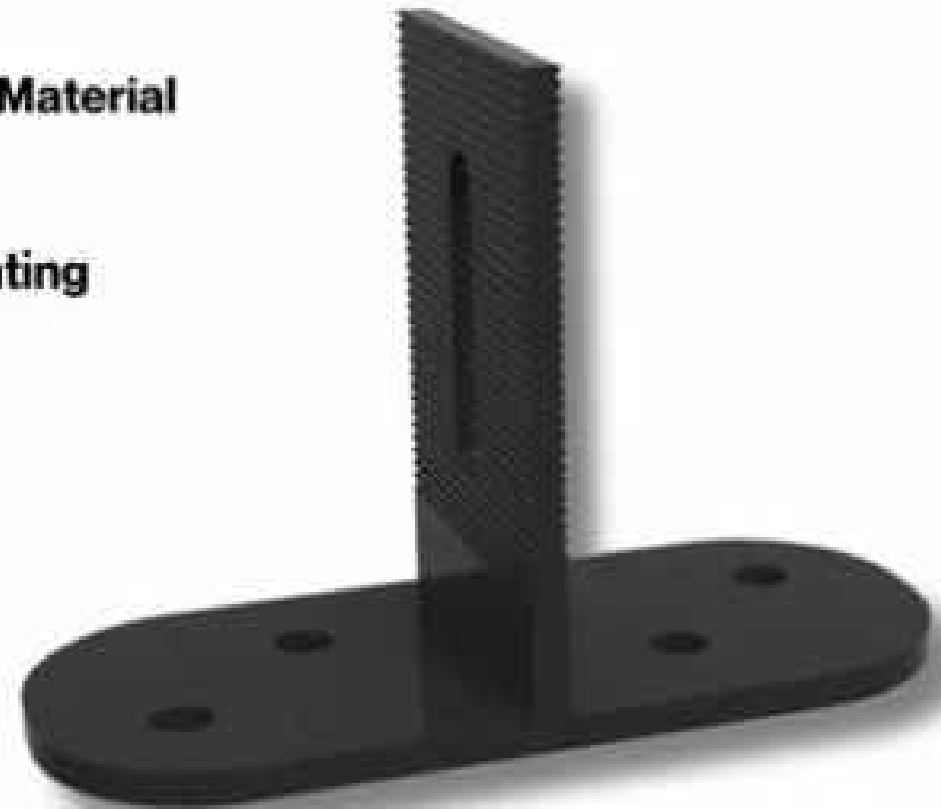
Peel-and-Place Installation



High-Quality Material



Secure Mounting



Q Roll over image to zoom in

Maple Leaf Solar PV Mounting Rapid Installation Black 7'8" Rail **Functionality**



Superior Weather Resistance



AS/NZS 1170 Certified



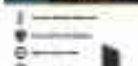
High-Quality Material



Secure Mounting



🔍 Roll over image to zoom in



Maple Leaf Black T-Foot with Butyl Seal **Functionality**



**Wind Load
36m/s**



**Snow Load
1.4KN/m2**



**No Sealant
Required**



**No Predrilling
Needed**



🔍 Roll over image to zoom in